Chapter Summary

Key Terms

3.1 Prime and Composite Numbers

- natural numbers
- factor of a number
- multiple of a number
- prime number
- composite number
- prime factorization
- greatest common divisor (GCD)
- least common multiple (LCM)

3.2 The Integers

- integer
- absolute value
- average of a set of numbers

3.3 Order of Operations

- order of operations
- PEMDAS

3.4 Rational Numbers

- density property of rational numbers
- improper fraction
- lowest terms
- mixed number
- rational number
- repeating decimal
- terminating decimal

3.5 Irrational Numbers

- conjugate numbers
- difference of squares
- irrational numbers
- lowest terms
- rationalize the denominator

3.6 Real Numbers

- complex number
- imaginary number
- real number

3.7 Clock Arithmetic

- clock arithmetic
- modulo 7
- modulo 12

3.8 Exponents

- base
- exponent

3.9 Scientific Notation

- scientific notation
- standard notation

3.10 Arithmetic Sequences

- sequence
- term of a sequence
- arithmetic sequence
- first term
- constant difference

3.11 Geometric Sequences

- geometric sequence
- common ratio

3.11 Geometric Sequences

- common ratio
- geometric sequence

Key Concepts

3.1 Prime and Composite Numbers

- The natural numbers can be categorized as 1, prime numbers, and composite numbers.
- Prime numbers have as their only factors 1 and themselves.
- Composite numbers have at least three distinct factors.
- Composite numbers can be written in their prime factorization form, which is found by repeatedly factoring prime factors from the number.
- The greatest common divisor (GCD) of a set of numbers is the largest integer that divides all of the numbers in the set. The prime factorizations of the numbers can be used to identify the greatest common divisor.
- The least common multiple (LCM) of a set of numbers is the smallest integer that is divisible by all of the numbers in the set. The prime factorizations of the numbers can be used to identify the least common multiple.
- There are various ways that the GCD and LCM are applied.

3.2 The Integers

- A set of numbers that can be built from the natural numbers are the integers, which consist of the natural numbers, zero (0), and the negatives of the natural numbers.
- Integers are often graphed on a number line, which helps display the relative positions and values of those numbers.
- The number line can be used to visualize when one integer is larger than or smaller than another integer.
- Arithmetic operations with integers are similar to the operations with natural numbers, except that the sign (positive or negative) of the numbers will determine the sign (positive or negative) of the result.

3.3 Order of Operations

- Establishing shared rules on which arithmetic operations are calculated first is necessary. Without them, different people may find different values for the same expression.
- The highest precedence is with expressions in parentheses. This allows parts of an expression to be calculated in an order different than the basic order of operations.
- The lowest precedence is addition and subtraction, as they are the basis for all other calculations.
- Multiplication and division have precedence over addition and subtraction, as they are representations of repeated addition or subtraction.
- Exponents have precedence over multiplication and division, as they represent repeated multiplication and division.

3.4 Rational Numbers

- Rational numbers are fractions of integers, and can always be written as an integer divided by an integer.
- The numerator and denominator of a fraction may have common factors. In such cases, the fraction can be reduced by canceling common factors. When the numerator and denominator of a fraction have no common factors, the fraction is said to be reduced.
- An improper fraction is one with a numerator larger than the denominator. Such a fraction can be rewritten as an integer plus a proper fraction. This is called a mixed number.
- Using division and remainder, an improper fraction may be written as a mixed number.
- A mixed number can be converted to an improper fraction by reversing the process for changing an improper fraction to a mixed number.

- The arithmetic operations or addition, subtraction, multiplication and division can all be performed on rational numbers.
- Addition and subtraction of rational numbers can be performed after a common denominator has been identified, and the fractions have been converted to forms having the common denominator.
- Multiplication and division of rational numbers can be performed without regard to common denominators.
- Between any two rational numbers, there is always another rational number. This is the density property of the rational numbers.

3.5 Irrational Numbers

- Irrational numbers are numbers that cannot be written as an integer divided by another integer. One example is pi, denoted π. Another collection of irrational numbers are natural numbers that are not perfect squares.
- Some irrational numbers can be written as a rational part multiplied by an irrational part. If two irrational numbers have the same irrational parts, they can be added or subtracted.
- When irrational numbers are similar, on can multiply and divide the numbers without a calculator.
- Since $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$, and $\sqrt{a} \div \sqrt{b} = \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$, products and quotients of square roots can be determined.
- Because $\sqrt{a^2} = a$ and $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$, it is possible to simplify square root expressions so the radicand contains no perfect square factors.
- When a fraction has an irrational number as its denominator, it is possible to convert the denominator into a rational number using its conjugate. Doing so involves multiplying the numerator and denominator by the conjugate of the denominator, and then applying the difference of squares formula.
- With a single square root term
- Using conjugate numbers for two term denominators

3.6 Real Numbers

- Real numbers is the collection of all rational and irrational numbers. Conceptually, it is the collection of all values that can be represented on a number line, or, as a length along with sign.
- The subsets of the real numbers include the natural numbers, integers, rational numbers and irrational numbers. The natural numbers are a subset of the integers, which is a subset of the rational numbers. The rational and irrational numbers are disjoint sets.
- The real numbers, due to order of operation rules and that performing arithmetic operations on real number always results in a real number, have arithmetic properties that apply in all cases. There include the distributive property, the commutative property, and the associative property. Also, every real number has an additive inverse and, except for zero (0), have a multiplicative inverse.

3.7 Clock Arithmetic

- Clock arithmetic uses the idea that after 12 o'clock comes 1 o'clock. For clock arithmetic, this means that every time 12 is passed in an arithmetic process, the next number is 1, not 13.
- To determine the clock result of an arithmetic operation, divide the final result by 12 and keep the remainder. If the remainder is 0, then the time is 12 o'clock.
- Clock arithmetic is technically called modulo 12 arithmetic. To perform modulo 12 arithmetic, calculate the expression, then divide the result by 12. The modulo 12 result is the remainder.
- Days, in our system, pass in groups of seven. To calculate in day arithmetic, modulo 7 is used. To perform modulo 7 arithmetic, calculate the expression, then divide the result by 7. The modulo 7 result is the remainder.

3.8 Exponents

- Exponents are used to express multiplying a number by itself a number of times. The number being multiplied by itself is the base. The number of times it is multiplied by itself is the exponent, which is often referred to as the power.
- Understanding that exponents represent repeated multiplication of a base makes it possible to establish some rules for combining exponential expressions, using the product rule, the quotient rule, and the power rule. Additionally, it allows us to formulate distributive rules for exponents.
- Any non-zero number raised to the 0th power is 1. This makes the definition of the 0th power consistent with the division rule for exponents.
- For consistency, negative exponents represent the reciprocal of the base raised to the power, so that $a^{-n} = \frac{1}{a^n}$, provided that $a \neq 0$.

3.9 Scientific Notation

- Some numbers are so large or so small that writing the number out is clumsy and make it difficult to determine the true size of the number. Scientific notation makes the number more readable and make the relative size of the number immediately apparent.
- A number written in scientific notation is a number at least 1 and smaller than 10 multiplied by 10 raised to an exponent. Converting between scientific notation and standard notation involves correctly applying multiplication and division by powers of 10, which in practice equates to understanding how moving the decimal point of a number impacts the exponent of 10.
- Adding and subtracting numbers in base 10 requires the exponent of 10 in each number be the same. Once the numbers are converted to have the same exponent with the ten, then the numbers are added or subtracted as indicated, with the power of 10 remaining the same. If the result is not in scientific notation (for instance, the number has exceeded 10), then then number must be converted into scientific notation.
- Multiplying and dividing numbers in scientific notation is done by multiplying or dividing the number parts, then multiplying or dividing the 10 raised to the power parts, then multiplying those two results. If the new number is not in scientific notation, then the result must be converted into scientific notation.

3.10 Arithmetic Sequences

- A sequence is a list of numbers. Any individual number in that list, or sequence, is a term of the sequence. A specific term of a sequence is denoted by the sequence symbol with a subscript indicating where the term in the sequence is.
- A special form of a sequence is an arithmetic sequence. Each arithmetic sequence is determined by its first term and its constant difference. Any term in an arithmetic sequence is determined by adding the constant difference to the preceding term.
- If the first term and the constant difference of an arithmetic sequence are known, then any term of the sequence can be found directly.
- Because arithmetic sequences follow such a strict pattern, the sum of the first n terms of an arithmetic sequence can be determined with the formula $s_n = n\left(\frac{a_1+a_n}{2}\right)$.

3.11 Geometric Sequences

- A special form of a sequence is a geometric sequence. Each geometric sequence is determined by its first term and its constant ratio. Any term in a geometric sequence is determined by multiplying the constant ratio to the preceding term.
- If the first term and the constant ratio of a geometric sequence are known, then any term of the sequence can be found directly.
- Because geometric sequences follow such a strict pattern, the sum of the first n terms of a geometric sequence can be determined with the formula $s_n = a_1\left(\frac{1-r^{n-1}}{1-r}\right)$.
- Finding the sum of a finite geometric sequence
- Applying arithmetic sequences

3.11 Geometric Sequences

- Geometric sequence.
- Finding an arbitrary term in a geometric sequence.
- Constant ratio.
- Finding the sum of a finite geometric sequence.
- Applying arithmetic sequences.

Videos

3.1 Prime and Composite Numbers

- Divisibility Rules (https://openstax.org/r/Divisibility_Rules)
- Illegal Prime Number (https://openstax.org/r/Illegal_Prime_Number)
- Using a Factor Tree to Find the Prime Factorization (https://openstax.org/r/Using_a_Factor_Tree_to_Find_the_Prime_Factorization)
- Finding the Prime Factorization of 168 (https://openstax.org/r/Finding_the_Prime_Factorization_of_168)
- Using Desmos to find the GCD (https://openstax.org/r/Using_Desmos_to_find_the_GCD)
- Applying the GCD (https://openstax.org/r/Applying_the_GCD)

- Finding the LCM (https://openstax.org/r/Finding_the_LCM)
- Using Desmos to find the LCM (https://openstax.org/r/Using_Desmos_to_find_the_LCM)
- Application of LCM (https://openstax.org/r/Application_of_LCM)

3.2 The Integers

- Graphing Integers on the Number Line (https://openstax.org/r/Graphing_Integers_on_the_Number_Line)
- Comparing Integers Using the Number Line (https://openstax.org/r/Comparing_Integers_Using_the_Number_Line)
- Evaluating the Absolute Value of an Integer (https://openstax.org/r/Evaluating_the_Absolute_Value_of-an-Integer)

3.3 Order of Operations

- Order of Operations 1 (https://openstax.org/r/Order_of_Operations_1)
- Order of Operations 2 (https://openstax.org/r/Order_of_Operations_2)
- Order of Operations 3 (https://openstax.org/r/Order_of_Operations_3)
- Order of Operations 4 (https://openstax.org/r/Order_of_Operations_4)

3.4 Rational Numbers

- Introduction to Fractions (https://openstax.org/r/Introduction_to_Fractions)
- Equivalent Fractions (https://openstax.org/r/Equivalent_Fractions)
- Reducing Fractions to Lowest Terms (https://openstax.org/r/Reducing_Fractions_to_Lowest_Terms)
- Using Desmos to Reduce a Fraction (https://openstax.org/r/Using_Desmos_to_Reduce_a_Fraction)
- Adding and Subtracting Fractions with Different Denominators (https://openstax.org/r/Adding_and_Subtracting_Fractions)
- Converting an Improper Fraction to a Mixed Number Using Desmos (https://openstax.org/r/Improper_Fraction_to_Mixed_Number)
- Multiplying Fractions (https://openstax.org/r/Multiplying_Fractions)
- Dividing Fractions (https://openstax.org/r/Dividing_Fractions)
- Order of Operations Using Fractions (https://openstax.org/r/Operations_Using_Fractions)
- Finding a Fraction of a Total (https://openstax.org/r/Finding_Fraction_of_Total)
- Converting Units (https://openstax.org/r/Converting_Units)

3.5 Irrational Numbers

- The Philosophy of the Pythagoreans (https://openstax.org/r/Philosophy_of_Pythagoreans)
- Using Desmos to Find the Square Root of a Number (https://openstax.org/r/square_root_of_a_number)
- Simplifying Square Roots (https://openstax.org/r/Simplifying_Square_Roots)
- Rationalizing the Denominator (https://openstax.org/r/Rationalizing_Denominator)

3.6 Real Numbers

- Properties of the Real Numbers 1 (https://openstax.org/r/Properties_of_the_Real_Numbers_1)
- Properties of the Real Numbers 2 (https://openstax.org/r/Properties_of_the_Real_Numbers_2)
- Properties of the Real Numbers 3 (https://openstax.org/r/Properties_of_the_Real_Numbers_3)

3.6 Real Numbers

- Arthur Benjamin TED talk, Faster than a Calculator (https://openstax.org/r/Arthur_Benjamin_TED_talk,_Faster_than_a_Calculator)
- Identifying Sets of Real Numbers (https://openstax.org/r/Sets_of_Real_Numbers)
- Properties of the Real Numbers #1 (https://openstax.org/r/Properties_of_the_Real_Numbers_1)
- Properties of the Real Numbers #2 (https://openstax.org/r/Properties_of_the_Real_Numbers_2)
- Properties of the Real Numbers #3 (https://openstax.org/r/Properties_of_the_Real_Numbers_3)

3.7 Clock Arithmetic

- Determining the Day of the Week for Any Date in History (https://openstax.org/r/Determining_the_Day_of_the_Week_for_Any_Date_in_History)
- Clock Arithmetic (https://openstax.org/r/Clock_Arithmetic)

3.8 Exponents

- Exponential Notation (https://openstax.org/r/Exponential_Notation)
- Product and Quotient Rule for Exponents (https://openstax.org/r/Product_and_Quotient_Rule_for_Exponents)
- Fraction Raised to a Power (https://openstax.org/r/Fraction_Raised_to_a_Power)

- Simplifying Expressions with Exponents (https://openstax.org/r/Simplifying_Expressions_with_Exponents)

3.9 Scientific Notation

- Converting from Standard Form to Scientific Notation Form (https://openstax.org/r/Converting_from_Standard_Form_to_Scientific_Notation_Form)
- Converting from Scientific Notation Form to Standard Form (https://openstax.org/r/Converting_from_Scientific_Notation_Form_to_Standard_Form)
- Multiplying Numbers in Scientific Notation (https://openstax.org/r/Multiplying_Numbers_in_Scientific_Notation)
- Dividing Numbers in Scientific Notation (https://openstax.org/r/Dividing_Numbers_in_Scientific_Notation)
- Application of Scientific Notation (https://openstax.org/r/Application_of_Scientific_Notation)

3.10 Arithmetic Sequences

- Arithmetic Sequences (https://openstax.org/r/Arithmetic_Sequences)
- Finding the First Term and Constant Difference for an Arithmetic Sequence (https://openstax.org/r/Finding_the_First_Term_and_Constant_Difference_for_an_Arithmetic_Sequence)
- Finding the Sum of a Finite Arithmetic Sequence (https://openstax.org/r/Finding_the_Sum_of_a_Finite_Arithmetic_Sequence)
- Fibonacci Sequence and “Lateralus” (https://openstax.org/r/Fibonacci_Sequence_and_Lateralus)

3.11 Geometric Sequences

- Geometric Sequences (https://openstax.org/r/Geometric_Sequences)
- Sum of a Finite Geometric Sequence (https://openstax.org/r/Sum_of_a_Finite_Geometric_Sequence)

Formula Review

3.4 Rational Numbers

$\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}$
$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$
$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$ $\frac{a}{b} \div \frac{c}{d} = \frac{a}{d} \times \frac{d}{c} = \frac{a \times d}{b \times c}$

3.5 Irrational Numbers

$$\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$$

$$a \times x \pm b \times x = (a \pm b) \times x$$

$$\sqrt{a} \div \sqrt{b} = \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

$$a^2 - b^2 = (a - b)(a + b)$$

3.6 Real Numbers

$a \times (b + c) = a \times b + a \times c$
$a + b = b + a$
$a \times b = b \times a$
$a + (b + c) = (a + b) + c$
$a \times (b \times c) = (a \times b) \times c$
$a + 0 = a$
$a \times 1 = a$
$a + (-a) = 0$
$a \times \left(\frac{1}{a}\right) = 1$

3.8 Exponents

$$a^n a^m = a^{n+m}$$

$$\frac{a^n}{a^m} = a^{(n-m)}$$

$a^0 = 1$, provided that $a \neq 0$

$$(a \times b)^n = a^n \times b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\left(a^n\right)^m = a^{(n \times m)}$$

$a^{-n} = \frac{1}{a^n}$, provided that $a \neq 0$

3.10 Arithmetic Sequences

$a_i = a_1 + d \times (i - 1)$

$d = \frac{a_j - a_i}{j - i}$

$a_1 = a_i - d\,(i - 1)$

$s_n = n\left(\frac{a_1 + a_n}{2}\right)$

3.11 Geometric Sequences

$a_n = a_1 r^{n-1}$

$s_n = a_1\left(\frac{1 - r^{n-1}}{1 - r}\right)$

3.11 Geometric Sequences

$a_n = a_1 r^{n-1}$

$s_n = a_1\left(\frac{1 - r^{n-1}}{1 - r}\right)$

Projects

Encryption began at least as far back as the Roman Empire. During the reign of Caesar, a particular cypher was used, fittingly named the Caesar Cypher. This encryption process granted the Romans a great tactical advantage. Even if a message was intercepted, it would not make sense to the person intercepting the message.

Find four instances when encryption was used and cracked over the course of history.

The Golden Ratio in Art and Architecture

The golden ratio has been used in art and architecture as far back as ancient Greece (possibly further). It also appears in South America (Incan architecture). Find five instances of the use of the golden ratio in art or architecture and describe its use in each of those instances.

Your Budget

Budgeting either is, or will shortly be, an important aspect of your life. Managing money well reduces stress in your life, and provides space for planning for future expenses, such as vacations or home improvements.

Imagine your life 10 years from now. Estimate your monthly income. Identify expenses you will encounter monthly (mortgage or rent, car payment, insurance, entertainment, etc.). Decide on an amount you plan to save monthly (this is treated as an expense). Create a spreadsheet with those values. Record your monthly net income (your income minus your expenses). Determine how much money you will have saved over the course of 5 years (ignore interest). Write a reflection on your anticipated financial health.

Estimating Pi

The value of pi is the ratio of the circumference of a circle to the diameter of the circle. It is also equal to the ratio of the area of the circle to the square of the radius of the square.

Research three ways to physically estimate pi.

Estimate pi using all three processes you found.

Present your process and solutions in class.

Design Your Own Shift Cypher

A cypher is a message written in such a way as to mask its contents. Changing a message into its cypher form is called encryption. Decryption or deciphering is the process of changing a cyphertext message back into the original (legible) message. One process of encryption is to scramble the letters, symbols, and punctuation of a message according to a mathematical rule. One rule that could be used for such a cypher is addition in a chosen modulus. In this project, you will create such a cypher, encrypt a message, and then decrypt the message.

Step 1: Choose the letters, symbols, and punctuation marks you want to allow in your messages. This should include at least the uppercase letters and a space character. This is your character set.

Step 2: Count the number of characters you will use. Label this number n.

Step 3: Pair each character of your character set an integer from 0 to $(n - 1)$. Do not assign more than one character to an integer.

Step 4: Choose an integer between 1 and $(n - 1)$. This will be the number used to create the cypher. Label this number s.

Step 5: Write a message using your character set.

Step 6: Replace every character in your message by the integer with which it was paired in Step 3.

Step 7: For every number, x, from Step 7, perform the addition $x + s$ (mod n).

Step 8: Replace every number found in Step 7 with the character with which it was paired in Step 3. This is your cyphertext.

To decrypt your cyphertext, reverse the steps above.

Step 1: Replace the cyphertext characters with the paired values.

Step 2: For each value x, perform the *subtraction* $x– s$ (mod n).

Step 3: Replace the numbers from Step 2 with their paired characters from the character set.

The message is then deciphered.

Design Your Own Cypher Using Multiplication

A cypher is a message written in such a way as to mask its contents. Changing a message into its cypher form is called encryption. Decryption or deciphering is the process of changing a cyphertext message back into the original (legible) message. One process of encryption is to scramble the letters, symbols, and punctuation of a message according to a mathematical rule. One rule that could be used for such a cypher is multiplication in a chosen modulus. In this project, you will create such a cypher, encrypt a message, then decrypt the message.

Step 1: Choose the letters, symbols, and punctuation marks you want to allow in your messages. This should include at least the uppercase letters and a space character. This is your character set.

Step 2: Count the number of characters you will use. Label this number n.

Step 3: Pair each character of your character set an integer from 0 to $(n - 1)$. Do not assign more than one character to an integer.

Step 4: Choose an integer, labeled s, between 1 and $(n - 1)$ so that $\text{GCF}(n, s) = 1$. This will be the number used to create the cypher.

Step 5: Write a message using your character set.

Step 6: Replace every character in your message by the integer with which it was paired in Step 3.

Step 7: For every number, x, from Step 6, perform the multiplication $x - s$ (mod n).

Step 8: Replace every number found in Step 7 with the character with which it was paired in Step 3. This is your cyphertext.

Before beginning to decrypt in this cypher, you need to know the **multiplicative inverse** of the value you chose as *s*.

Step 1: The multiplicative inverse of s is the number that, when multiplied by s in your modulus, equals 1. To find this, you will have to multiply s and every number between 2 and $(n– 1)$ until the product is 1 (mod n). Once this number is found, the message can be decrypted. Call this number r.

Step 2: To decrypt your cyphertext, replace the cyphertext characters with the paired values.

Step 3: For each of the value, x, perform the multiplication $x \times r(\text{mod}\, n)$.

Step 4: Replace the numbers from Step 3 with their paired characters from the character set.

The message is then deciphered.

Chapter Review

Prime and Composite Numbers

1. Identify which of the following numbers are prime, composite, or neither: 201, 34, 17, 1, 37.
2. Find the prime factorization of 500.
3. Find the greatest common divisor of 80 and 340.
4. Find the greatest common divisor of 30, 40, and 70.
5. Find the least common multiple of 45 and 60.
6. Bella and JJ volunteer at the zoo. Bella volunteers every 8 days, while JJ volunteers every 14 days. How many days pass between days they volunteer together?

The Integers

7. Identify all the integers in the following list: $-4, \frac{2}{7}, 15, 97.5, 0, \sqrt{122}$.
8. Plot the following on the same number line: $1, 5, -2, 0$.
9. What two numbers have absolute value of 18?
10. Calculate $14 - (-12)$.
11. Calculate $(-19) \times 4$.
12. Calculate $(-240) \div (-12)$.
13. Six students rent a house together. The total monthly rent (including heat and electricity) is $3,120. If they all pay an equal amount, how much does each student pay?

Order of Operations

14. Calculate $5 \times 3 - (-12)$.
15. Calculate $4^2 \times 6 + 2^3$.
16. $(4 - 6) \times 2$
17. $(8 - 1)^3 - 3 \times 9$
18. $120 - (51 + 12) \div 3^2$

Rational Numbers

19. Reduce $\frac{90}{153}$ to lowest terms.
20. Convert $\frac{430}{25}$ to a mixed number and reduce to lowest terms.
21. Convert $\frac{9}{5}$ to decimal form.
22. Convert $\frac{5}{11}$ to decimal form.
23. Calculate $\frac{4}{15} + \frac{9}{10}$ and reduce to lowest terms.
24. Compute $\frac{3}{10} \div \frac{12}{25}$ and reduce to lowest terms.
25. Determine 30% of 400.
26. 18 is what percent of 40?
27. In Professor Finnegan's Science Fiction course, there are 60 students. Of those, 15% say they've read *A Hitchhiker's Guide to the Galaxy*. How many of the students have read that book?

Irrational Numbers

28. Simplify the square root by expressing it in lowest terms: $\sqrt{275}$.
29. Calculate $4\sqrt{13} - 15\sqrt{13}$ without a calculator. If not possible, explain why.

30. Calculate $\left(2.3\sqrt{5}\right) \times \left(4.2\sqrt{3}\right)$ without a calculator. If not possible, explain why.

31. Rationalize the denominator of $\frac{6}{\sqrt{22}}$, and simplify the fraction.

32. Find the conjugate of $4\sqrt{7} + 3$ and find the product of $4\sqrt{7} + 3$ and its conjugate.

33. Rationalize the denominator of $\frac{3}{8 + \sqrt{34}}$ and simplify the fraction.

Real Numbers

34. Identify the numbers of the following list as a natural number, an integer, a rational number, or a real number: $\sqrt{37}$, −0.43, 18, −43, 12π.

35. Identify the property of real numbers illustrated here: $14 + 19 = 19 + 14$.

36. Identify the property of real numbers illustrated here: $87.4 + 0 = 87.4$.

37. Identify the property of real numbers illustrated here: $2 \times (\sqrt{31} - 12) = 2\sqrt{31} - 2 \times 12$.

38. Identify the property of real numbers illustrated here: $98 \times 41 = 41 \times 98$.

39. Use mental math to calculate 21×99.

Clock Arithmetic

40. Determine 74 modulo 9.

41. Use clock arithmetic to calculate $13 + 25$.

42. Use clock arithmetic to calculate 4×8.

43. It is Wednesday. What day of the week will it be in 44 days?

44. It is 4:00. What time will it be in 100 hours?

45. Security guards with Acuriguard submit a report on campus activity every 4 days. If they make a report on a Monday, what day of the week will it be after 10 more reports?

Exponents

46. Use exponent rules to simplify $17^8 \times 17^3$.

47. Use exponent rules to simplify $\frac{15^9}{15^{-5}}$.

48. Use exponent rules to simplify $(3k)^6$.

49. Use exponent rules to simplify $\left(\frac{y}{4}\right)^{11}$.

50. Use exponent rules to simplify $\left(51^3\right)^6$.

51. Use exponent rules to simplify $\left(\frac{3x^4}{5}\right)^7$.

52. Rewrite $\frac{5^2}{x^4}$ without a denominator.

53. Rewrite $16z^{-9}$ without negative exponents.

Scientific Notation

54. Convert 0.0000452 to scientific notation.

55. Convert 3.01×10^5 to standard notation.

56. Convert 42.9×10^{-9} to scientific notation.

57. Calculate $4.51 \times 10^5 - 9.11 \times 10^4$.

58. Calculate $\left(9.15 \times 10^3\right) \div \left(3 \times 10^8\right)$.

59. The Sextans Dwarf Spheroidal Galaxy has diameter 8,400 light years (ly). Express this in Scientific notation.

60. The Reticulum II Galaxy has diameter 3.78×10^2 light years (ly), while the Andromeda Galaxy has a diameter of

2×10^5. How many times bigger is the Andromeda Galaxy compared to the Reticulum II Galaxy?

Arithmetic Sequences

61. Determine the common difference of the following sequence: {19, 13, 7, 1, -5, ...}.

62. Find the 25th term, a_{25}, of the arithmetic sequence with $a_1 = 13$ and $d = 1.7$.

63. Find the first term and the common difference of the arithmetic sequence with 8th term $b_8 = 50$ and 15th term $b_{15} = 106$.

64. Find the sum of the first 30 terms, s_{30}, for the arithmetic sequence with first term $a_1 = -4$ and common difference $d = 3.5$.

65. Jem makes a stack of 5 pennies. Each day, Jem adds three pennies to the stack. How many pennies are in the stack after 10 days?

Geometric Sequences

66. Determine the common ratio of the following geometric sequence: {6, 18, 54, 162, ...}.

67. Find the 6th term, c_6, of the geometric sequence with $c_1 = 400$ and $r = 0.25$.

68. Find the sum of the first 12 terms, s_{12}, for the geometric sequence with first term $a_1 = -4$ and common ratio $r = -1.25$.

69. Carolann and Tyler deposit $8,500 in an account bearing 5.5% interest compounded yearly. If they do not deposit any more money in that account, how much will be in the account after 15 years?

70. The total number of ebooks sold in 2013 was 242 million ($a_1 = 242$). Each year, the number of ebooks sold has declined by 3% ($r = 0.97$). How many ebooks were sold between 2013 and 2022?

Chapter Test

1. Find the prime factorization of 300.

2. Tiles will be used to cover an area that is 650 cm × 1,200 cm. What is the largest size square tile that can be used so that all the tiles used are full tiles?

3. Calculate the following: $3 \times \left(\frac{4}{3} \times (14 - 8) + 3\right) - 4 \times \left(8 - 2^2\right)$.

4. What does PEMDAS stand for?

5. David's tax return is for $1,560. He decides to spend 20% of that return. How much does David spend?

6. Convert $\frac{305}{26}$ into a mixed number.

7. Convert $0.\overline{659}$ to a fraction of integers.

8. Simplify the following square root: $\sqrt{6237}$.

9. What is the conjugate of $4 - \sqrt{10}$?

10. Rationalize the denominator of $\frac{2}{10 + \sqrt{13}}$.

11. What two sets of numbers comprise the real numbers?

12. Which property of the real numbers is shown: $4\,(a + 7) = 4a + 4 \times 7$?

13. Georita believes it will take 30 hours for her and her family to drive to their vacation. If they leave at 2:00, what time should they arrive (ignore AM/PM)?

14. Suppose a bacterium in the gut has a generation time (time to divide) of 16 hours. If it first divides at 4:00, what time will it be when they divide the 80th time afterward?

15. Simplify $\left(\frac{9x}{a^2}\right)^5$.

16. The moon is 3.844×10^8 m from Earth. A dollar bill has a thickness of 1.0922×10^{-4} m. If dollar bills could be stacked perfectly, how many would it take to reach the moon?

17. A single long table can seat 8 people, 3 on each side and 1 on each end. If a second table is added to the first, end to end, then 14 people can sit at the table, 6 per side and 1 at each end. Adding another table adds another 6 people. How many people can sit at a table made by placing 10 of these tables end to end?

18. The first row of a theater seats 25 people. Each following row seats 2 more people. If there are 80 rows in the theater, how many people, total, can sit in the theater?

19. Alice deposits $2,500 in a bond yielding 6% interest compounded annually. How much is the bond worth in 20 years?

20. What is the 15th term of a geometric sequence with first term 5 and common ratio 3?